

KinTCN: A Temporal Convolutional Network for Fitness-to-Drive Scoring from Vehicle Kinematics

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Abstract

Distracted and cognitively impaired driving remains a leading cause of road traffic fatalities, yet existing detection systems either require intrusive physiological sensors, raise privacy concerns through camera monitoring, or are evaluated within-subject in ways that overestimate real-world performance. We argue that the correct problem formulation is not event prediction, i.e., forecasting whether a driver will make an error in the next N seconds, but continuous impairment state scoring: quantifying whether the driver is currently in a state associated with elevated error risk. We present **KinTCN**, a Temporal Convolutional Network that operationalises this framing into a **fitness-to-drive index** derived exclusively from four kinematic signals available in any modern vehicle: speed, steering wheel angle, steering torque, and lateral acceleration. Trained with weak supervision from severity-weighted safety-critical event annotations and evaluated under Leave-One-Participant-Out Cross-Validation (LOPO-CV) with strict anti-leakage normalisation, KinTCN achieves an AUROC of 0.81 [0.76, 0.86] and an Expected Calibration Error of 0.037 after Platt scaling across 18 evaluable participants ($N = 40$ total), representing gains of +0.071 in discrimination and 75% in calibration quality over the best recurrent baseline. A dedicated false positive audit on the remaining 22 participants confirms that the calibrated index stays well below the alert threshold during safe driving (group mean impairment probability 0.054), demonstrating selective response to genuine impairment patterns rather than arbitrary kinematic deviations. The broader contribution is a demonstration that cross-participant generalisation and probabilistic calibration are jointly necessary conditions for a deployable fitness-to-drive index, properties that within-subject evaluation protocols systematically obscure. KinTCN and the labelled dataset are publicly available.

1 Introduction

Road traffic injuries constitute one of the leading causes of preventable death worldwide. According to the World Health Organization, approximately **1.19 million people** die each year as a result of road traffic crashes [33]. In the

United States alone, the National Highway Traffic Safety Administration reported that distraction-affected crashes accounted for **3308 fatalities** in 2023, representing roughly 8% of all traffic deaths that year [22]. Despite decades of road safety improvements, distracted and cognitively impaired driving remains a persistent and growing threat, exacerbated by the proliferation of in-vehicle infotainment systems and mobile devices [36, 13].

Existing approaches to driver impairment detection fall broadly into four categories, each carrying fundamental practical limitations. *Physiological monitoring*, relying on heart rate, electrodermal activity, or electroencephalography, offers direct access to the driver’s arousal state but requires intrusive wearable or embedded sensors that are costly, susceptible to motion artefacts, and difficult to deploy at scale [11]. *Camera-based systems* that track eye gaze, head pose, or facial expression achieve strong performance in controlled laboratory settings but raise privacy concerns and degrade under variable lighting and partial occlusion [31]. A third line of work frames the problem as *crash or error prediction*, forecasting discrete safety-critical events from sensor streams. This formulation is, however, fundamentally ill-posed: errors are rare and highly context-dependent, and their occurrence is jointly determined by the driver’s state, road geometry, and surrounding traffic, factors that in-vehicle sensors alone cannot fully observe [6]. Naturalistic driving data confirm that driver-related factors are present in approximately 88% of crashes, yet no deployed system reliably forecasts the *next* error; what remains feasible is detecting the driver *state* that elevates error probability [6]. A fourth line of work monitors *vehicle kinematics* directly, using speed, steering, and lateral dynamics as proxies for driver impairment state. This approach requires no additional hardware beyond sensors already present in modern vehicles, yet existing kinematic-based systems either rely on hand-crafted thresholds or are evaluated within-subject in ways that overestimate real-world performance [13].

This distinction is not merely semantic, and kinematics offer a principled path forward. A well-established body of work in driving psychology demonstrates that vehicle kinematics carry reliable information about driver workload and attentional state. Steering entropy theory, introduced by Nakayama et al. [21] and further developed by Boer [4], formalises this intuition: under increased cognitive load, steering corrections become less predictable and more irregular, reflecting degraded closed-loop vehicle control. Subsequent work has confirmed that steering wheel reversal rates are sensitive to cognitive secondary task load [15], that cognitive distraction produces measurable degradation in speed maintenance and braking response [28, 27], and that visual and cognitive distraction produce distinct, identifiable signatures across speed regulation, lane-keeping precision, and lateral acceleration [7]. A comprehensive theoretical account of these effects is provided by Engström et al. [8], who establish that cognitive load systematically reduces the bandwidth of the driver’s closed-loop vehicle control, producing observable kinematic consequences before overt errors occur. Taken together, this body of evidence establishes that in-vehicle kinematic signals carry systematic information about driver impairment state, a finding grounded in decades of human factors research [21, 4, 8].

Yet this established link has not been operationalised into what practitioners actually need: a **fitness-to-drive index**, i.e., a continuous, real-time score that quantifies how safely a driver is currently operating, without requiring physiological sensors, cameras, or knowledge of the road environment. We argue that the correct framing is not “*will this driver make an error in the next N seconds?*”, a question that is ill-posed given the rarity and context-dependence of errors [6], but rather: “*is this driver currently in a state associated with elevated error risk?*” The former demands event prediction; the latter demands **impairment state scoring**. Existing kinematic-based systems fail to deliver this index in practice because they rely on hand-crafted features that do not generalise across the large inter-individual variability in baseline driving style, or because they are evaluated within-subject in ways that conflate individual style recognition with genuine impairment detection [13].

We present **KinTCN**, a Temporal Convolutional Network [3] trained on four standard in-vehicle kinematic signals, i.e., speed, steering wheel angle, steering torque, and lateral acceleration, collected during a controlled driving simulator study ($N = 40$ participants). Safety-critical error events serve as *weak supervision*: windows preceding an error are used to define elevated-risk periods and the model output is interpreted not as an error probability, but as a continuous **fitness-to-drive index**, i.e., a well-calibrated score that ranks impaired driving periods above safe ones. Evaluation is conducted under **Leave-One-Participant-Out Cross-Validation (LOPO-CV)**, the strictest available cross-participant protocol, ensuring that the index generalises to drivers never seen during training.

The contributions of this work are as follows:

- **A non-invasive fitness-to-drive index.** A continuous, real-time impairment score derived exclusively from four kinematic signals already available in modern vehicles, requiring no physiological sensors, cameras, or knowledge of the road environment.
- **KinTCN: a calibrated, cross-participant impairment scorer.** A Temporal Convolutional Network trained with weak supervision, focal loss, and SMOTE, evaluated exclusively under LOPO-CV with anti-leakage normalisation, and producing a calibrated fitness-to-drive index from four standard kinematic signals. The broader methodological contribution is a demonstration that cross-participant generalisation and probabilistic calibration are jointly necessary conditions for a deployable fitness-to-drive index, properties that within-subject evaluation protocols systematically obscure. KinTCN code is publicly available at [1].
- **A labelled driving dataset.** Forty participants, four kinematic signals, and severity-weighted safety-critical event annotations across different incident types. The dataset supports a dual-cohort evaluation design: 18 participants with sufficient incident rates for AUROC estimation, and 22 evaluated under a dedicated false positive audit, the safety-relevant ques-

tion for drivers whose incident rate is too low to support discrimination estimation. The dataset is publicly available at [1].

The remainder of the paper is organised as follows. Section 2 reviews related work on driver impairment detection and kinematic-based driver monitoring. Section 3 describes the study and dataset. Section 4 details the KinTCN method and evaluation protocol. Section 5 presents experimental results. Section 6 discusses findings and limitations. Section 7 concludes the work.

2 Related Work

Driver impairment detection sits at the intersection of driving psychology, signal processing, and machine learning. We review the literature along three axes: the theoretical and empirical basis for using vehicle kinematics as a proxy for driver impairment state; existing signal-based detection approaches and their practical limitations; and the methodological concern of cross-participant generalisation, which motivates the evaluation protocol adopted in this work.

Kinematic signatures of driver impairment. The link between vehicle kinematics and driver cognitive state has a firm theoretical and empirical basis. Steering entropy theory, introduced by Nakayama et al. [21] and formalised by Boer [4], establishes that degraded closed-loop vehicle control under cognitive load manifests as increased irregularity in steering corrections. Empirical support is extensive: Strayer and Johnston showed that even hands-free phone conversation impairs driving relative to an undistracted baseline [29]; Strayer et al. further demonstrated that cell-phone use produces speed and following-distance degradation comparable to legal alcohol impairment [28, 27]; Recarte and Nunes confirmed that mental workload reduces the attentional bandwidth available for vehicle control independently of visual distraction [26]; and Patten et al. showed that even hands-free mobile phone use imposes measurable cognitive costs on primary driving performance [23]. Engström et al. showed that visual and cognitive distraction produce distinct, identifiable signatures across speed regulation, lane-keeping precision, and lateral acceleration [7]; Kountouriotis et al. confirmed that steering wheel reversal rates are selectively sensitive to cognitive secondary task load even when gaze remains on the road [15]; and Klauer et al. demonstrated, in the landmark 100-Car Naturalistic Driving Study, that inattention is the primary contributing factor in near-crash events [14]. A comprehensive theoretical account is provided by Engström et al. [8], who establish that cognitive load systematically reduces the bandwidth of the driver’s closed-loop vehicle control, producing observable kinematic consequences before overt errors occur. More recently, Zheng et al. showed that increases in steering entropy predict the onset of performance degradation in a driving simulator [38]. Taken together, this body of evidence motivates the use of kinematic signals as a proxy for impairment state rather than as predictors of discrete error events.

Signal-based driver distraction detection. Leveraging these kinematic signatures for automatic detection has attracted sustained interest. Li et al. demonstrated that driving performance indicators derived from steering, speed, and lateral position in a large-scale naturalistic dataset can discriminate distracted from attentive driving with meaningful accuracy, highlighting steering entropy as the most informative single feature [19]. Kaplan et al. provide a comprehensive survey of signal-based driver behaviour analysis, noting that threshold-based and handcrafted-feature approaches achieve reasonable within-subject performance but fail to generalise across drivers due to large inter-individual variability in baseline kinematics [13]. Dingus et al. analysed the SHRP2 naturalistic driving study and confirmed that driver-related factors are implicated in approximately 88% of crashes, reinforcing the value of continuous driver state monitoring, whilst also demonstrating the rarity and context-dependence of discrete crash events, which limits the viability of an event-prediction framing [6].

Deep learning for driver state monitoring and the cross-participant generalisation gap. Classical machine learning approaches (Support Vector Machines, Random Forests, and Gradient Boosted Trees) have been widely applied to driver state classification from vehicle signals, typically operating on hand-engineered time-domain and frequency-domain features [13, 16, 35]. Xue et al. directly compared SVM and Random Forest on vehicle control signals for cognitive distraction detection, finding competitive accuracy within-subject, but neither model was evaluated across participants [35]. Deep learning methods, particularly Long Short-Term Memory networks [12] and CNN-LSTM hybrids [17], have emerged as the dominant paradigm for end-to-end learning from raw or lightly preprocessed driving time series. Wang et al. demonstrated that a Bi-LSTM with attention trained on vehicle dynamics alone achieves strong distraction detection performance on naturalistic driving data [32], reinforcing the viability of kinematic-only deep learning whilst again relying on within-subject evaluation. Temporal Convolutional Networks (TCNs), introduced by Bai et al. [3] as a competitive sequence modelling alternative to recurrent architectures, combine causal dilated convolutions [30] with residual connections [10] to achieve receptive fields that scale exponentially with depth, enabling efficient capture of long-range temporal dependencies. Despite their demonstrated advantages in generic sequence modelling [3], TCNs remain underexplored in the driver monitoring literature. The closest prior work using a TCN on vehicle dynamics is Zhao et al. [37], who apply a soft-thresholding TCN to abnormal driving behaviour recognition, reporting improved efficiency over recurrent baselines on similar kinematic inputs. We do not benchmark directly against their architecture for a reason that is specific rather than evasive: their model assigns windows to a fixed manoeuvre taxonomy (harsh braking, swerving) and produces a hard categorical label, whereas KinTCN produces a continuous, calibrated impairment probability that is agnostic to which manoeuvre is being executed. These differences in label space and output type mean that a head-

to-head AUROC comparison would measure how well each model solves its own task on its own labels — not which architecture better captures the driver’s impairment state. Beyond the task mismatch, Zhao et al. evaluate entirely within-subject, so their reported performance cannot be attributed to cross-participant generalisation. For these reasons, we treat their work as evidence that TCNs are competitive on vehicle-dynamics tasks, and not as a baseline against which our impairment index should be scored.

A persistent methodological concern underlying all of the above work is the reliance on within-subject evaluation protocols, in which training and test windows are drawn from the same participants [13]. This practice systematically inflates reported performance, because models can exploit individual-specific driving style rather than learning generalisable impairment signatures. The problem is compounded when studies perform data augmentation or normalisation using information from the test participant, a form of leakage that further overstates generalisation. A further limitation shared across this literature is the absence of output calibration: reported metrics are typically limited to classification accuracy or AUROC, with no assessment of whether predicted scores are probabilistically meaningful, a prerequisite for any deployable risk index. Leave-One-Participant-Out Cross-Validation (LOPO-CV) is the strictest available protocol for addressing the generalisation gap: each participant’s data are held out entirely, and all preprocessing and model fitting is performed exclusively on the remaining participants [13].

3 Dataset and Experimental Setup

[[TBC.]] The KinTCN model presented in this work was developed and evaluated using the vehicle telemetry and safety-related driving event data extracted from a larger multimodal driving dataset collected within the European DistriMuSe project¹. The complete dataset, which additionally includes driver-related and environmental data to support a broader range of research objectives [25], is described in a companion publication [[TBD]]. In the present work, only the vehicle- and safety performance - related data were considered, as they are sufficient for the proposed fitness-to-drive estimation approach. The vehicle-related dataset comprises conventional vehicle telemetry signals (e.g., vehicle speed, longitudinal and lateral acceleration, steering wheel angle, brake and accelerator pedal positions, yaw rate, lane position, and other vehicle dynamics measurements) together with automatically recorded safety-related driving events. These include collisions, red-light violations, speed violations, harsh braking events, emergency braking leading to a full stop, abrupt steering manoeuvres, and lane departure events, which provide objective indicators of driving performance and safety exploiting only vehicle data. The dataset was collected in July 2025 using a fixed-base driving simulator of one of the DistriMuSe partner (RE:LAB in Reggio Emilia, Italy) based on the SCANeR simulation platform, which reproduces realistic urban driving conditions while providing

¹<https://distrimuse.eu/>

full experimental control over traffic events and environmental variables. The simulator ensured that all participants experienced identical driving scenarios under reproducible conditions. The driving simulation comprised five categories of interactions with Vulnerable Road Users (VRUs), selected to represent common urban situations in which timely driver perception and reaction are essential for safe driving. These included, for example, pedestrians suddenly emerging from behind visual occlusions, pedestrians crossing against a red traffic light, and VRUs unexpectedly changing their direction while crossing the roadway. To preserve ecological validity while minimizing expectancy effects and strategic adaptation to the experimental events, the order of the scenarios was randomized across two alternative driving routes. The sequence was additionally constrained so that no more than three safety-critical events could occur consecutively. Each participant completed two driving sessions under two different driver-state conditions. In the non-distracted driving condition, participants performed only the primary driving task. In the distracted driving condition, participants simultaneously performed six visual search tasks per minute on a secondary display, thereby inducing a controlled level of visual-cognitive distraction. To control for potential order, learning, fatigue, and carry-over effects, both the driving condition and the route sequence were counterbalanced according to a balanced experimental design comprising four experimental groups. Participants were allocated to the groups to ensure an approximately equal number of subjects in each condition combination. The data collection campaign involved 41 participants (15 female, 26 male) with a mean age of 33.2 years ($SD = 8.4$). Regarding driving experience, 3 participants reported less than 3 years of driving, 16 had between 3 and 10 years, and 22 had more than 10 years. The data collection campaign was conducted in accordance with a protocol approved by the Transdisciplinary Ethics Committee of the University of Suor Orsola Benincasa). Before taking part in the driving sessions, all participants received detailed information about the experimental procedures and provided written informed consent.

- participants: OK - Signals recorded and sampling rate: TBC - Error types logged: OK

4 The KinTCN Fitness-to-Drive Scorer

Computing a fitness-to-drive index requires a model that can extract temporal patterns from multivariate driver time series in real time, generalise across drivers with different baseline styles, and produce a well-calibrated continuous score rather than a hard binary decision. Temporal Convolutional Networks (TCNs) [3] satisfy all three requirements: they enforce causality by construction, so that the output at time t depends only on inputs at times $\leq t$; they use dilated convolutions [30] to capture dependencies spanning hundreds of timesteps without recurrence; and they are trained discriminatively end-to-end.

We propose **KinTCN**, a TCN trained on four standard in-vehicle kinematic signals: speed, steering wheel angle, steering torque, and lateral acceleration.

The network outputs a calibrated impairment probability $\hat{p} \in [0, 1]$; the fitness-to-drive index is its complement, $f = 1 - \hat{p}$, so that f close to 1 indicates a driver in full control and f close to 0 indicates elevated impairment risk. The true network target, the driver’s impairment state, is a latent variable that cannot be observed directly. What can be observed are safety-critical error events, such as collisions and red-light violations, recorded during the study. KinTCN uses these events as *weak supervision*: a kinematic window is labelled as elevated-risk if an error is recorded in the near future, and as safe otherwise. Such supervision is weak because the label is a noisy proxy: an impaired driver may not make an error in a given window, and an attentive driver may make one due to road conditions; it is, however, the only supervision signal available without intrusive physiological measurement. The model therefore learns to score the driver’s control state, not to predict specific error events. Three sources of fragility common in driver monitoring research receive explicit attention in the design: data leakage across participants during preprocessing, class imbalance between rare error-preceding windows and abundant safe windows, and overoptimistic evaluation on participants seen during training. Each of these is addressed explicitly, and the design choices are detailed in the subsections that follow.

4.1 Label Derivation and Predictive Window

The model operates on fixed-length kinematic windows and must assign each one a scalar impairment probability before the next window arrives. Because the driver’s true impairment state is latent, each window is labelled through the safety-critical events that follow it: a window ending at time t is paired with the future interval $[t + G, t + G + H]$, and the events occurring within that interval constitute its weak supervision signal. The gap G enforces temporal separation between the kinematic observation and the labelled period, preventing the model from exploiting signatures that are contemporaneous with an error rather than significantly preceding it. The horizon H defines the width of the future period over which evidence is aggregated; a wider horizon increases label coverage at the cost of temporal precision. Both values are chosen empirically and their effect on performance is examined in Section 5.

To reflect the differential safety relevance of each event, the dataset records four types of safety-critical incident, each assigned a severity weight: Collision ($w = 5$), Red-light violation ($w = 3$), Panic braking with stop ($w = 2$), and Panic braking ($w = 1$). The composite risk score for a window is the sum of severity weights of the distinct event types that occur at least once in the associated future interval. A window is labelled positive ($y = 1$) if its score is ≥ 1 , i.e., if any safety-critical event occurs in the future interval, and negative ($y = 0$) otherwise. Higher-severity events increase the score without excluding lower-severity ones from the positive class; the score is later used to up-weight high-severity windows in the loss function.

Since safety-critical incidents are rare relative to normal driving, the labelling procedure inevitably produces a heavily skewed class distribution: positive windows constitute a small minority of the full window set. This imbalance is

structural rather than an artefact of the threshold choice: even at the minimum score threshold of 1, which maximises positive coverage by counting any incident as sufficient, error-preceding windows remain far outnumbered by safe ones. A model trained on the raw distribution has little incentive to attend to the minority class; it can minimise loss by assigning uniformly low impairment scores, achieving high accuracy whilst being useless as a risk detector. Correcting for this at both the data level and the loss level is therefore a prerequisite for meaningful training.

4.2 Preprocessing and Feature Engineering

Each driver exhibits a characteristic baseline style, e.g., habitual cruising speed, typical steering amplitude, preferred torque range, that would dominate any learned cross-participant representation if left uncontrolled. Each kinematic signal is therefore z-scored per driving session, i.e., standardised to zero mean and unit variance, using the first $L + G$ seconds of the session as the personal reference baseline, where L denotes the lookback duration, i.e., the length of the kinematic history fed to the network at each prediction step. This initial segment precedes the distraction induction and thus provides a distraction-free snapshot of the driver’s individual operating point. Extreme values are clipped at ± 3 standard deviations to suppress artefacts from rare transient manoeuvres. Because the model is evaluated under Leave-One-Participant-Out Cross-Validation (LOPO-CV), these normalisation statistics must be recomputed in each fold using only the training participants: allowing any statistic derived from the held-out driver to influence preprocessing would introduce leakage and produce overoptimistic generalisation estimates.

From each normalised recording, fixed-length windows of L seconds are extracted with a uniform stride of 5 s (consecutive windows therefore overlap by $L - 5$ seconds), converting the continuous time series into a sequence of $(L \times 4)$ signal matrices. Any window that extends beyond the end of the recording, or whose associated future labelling interval $[t + G, t + G + H]$ is not fully contained within the session, is discarded. The lookback duration L controls how much temporal context the network can attend to and, together with H , is treated as a hyperparameter whose effect on performance is examined in Section 5. Notice that, G is not treated as a hyperparameter. KinTCN enforces strict causality by construction: causal zero-padding ensures that the convolution at time t has access only to inputs at times $\leq t$, so no signal contemporaneous with or following an error can enter the prediction. Given this architectural guarantee, the sole function of G is to push the labelling interval beyond the observation window, not to prevent direct exploitation of error-concurrent kinematics. Any $G > 0$ therefore provides the required separation; setting G equal to the window stride of 5 s retains the maximum label coverage while satisfying this condition. Larger values yield no further benefit while reducing the number of valid labelled windows per session. G is therefore fixed at 5 s throughout experiments.

The raw four-channel window provides the network with only instantaneous signal values, giving it no direct access to how the signals are evol-

ing or how they deviate from local norms. Each window is therefore transformed into a richer representation: for each of the four kinematic signals, at every timestep, we compute the raw normalised value, the first-order temporal difference, a within-window z-score (capturing deviation from the window’s own mean, complementary to the session-level normalisation), and rolling mean and standard deviation at three temporal scales of 5s, 10s, and 25s. The multi-scale rolling statistics are motivated by the heterogeneous temporal structure of the target events: the 5s scale responds to abrupt kinematic changes characteristic of panic braking precursors, whereas the 25s scale captures the gradual degradation of vehicle control that tends to precede collisions and red-light violations. The resulting feature tensor has shape $(L \times 36)$ per window ($4 \text{ signals} \times 9 \text{ features per signal}$).

4.3 Model Design and Training

A central design requirement for KinTCN is strict causality: at any moment t , the impairment score may only depend on signals observed up to and including t , never on future measurements. KinTCN is a Temporal Convolutional Network [3] that enforces this by construction through one-sided, causal dilated convolutions that look only backwards in time. The architecture comprises five residual blocks with exponentially increasing dilations $d \in \{1, 2, 4, 8, 16\}$, yielding a receptive field of $1 + 2 \sum_d d = 63$ timesteps — sufficient to cover the entire $L=45\text{s}$ lookback window at 1 Hz. Each residual block applies causal zero-padding of $2d$ steps, followed by a 1×3 dilated convolution, Group Normalisation [34], ReLU activation, and a 10% channel dropout; a 1×1 projection convolution provides the residual shortcut when the channel dimension changes. The channel progression is $36 \rightarrow 32 \rightarrow 64 \rightarrow 64 \rightarrow 64 \rightarrow 64$, where 36 is the number of engineered features per timestep. The initial reduction from 36 to 32 channels projects the engineered feature space into a compact learned representation before expansion, reducing the risk of overfitting to redundant feature combinations in the first convolutional layer. Group Normalisation is preferred over Batch Normalisation because, under LOPO-CV, effective per-fold batch sizes are small and batch statistics become unreliable; Group Normalisation operates independently of batch size and remains stable across folds [34].

Not all moments within a window are equally informative: a brief episode of erratic steering may be far more diagnostic than the surrounding normal driving, yet a global average would dilute it. After the convolutional stack, KinTCN therefore applies a lightweight additive temporal attention mechanism [2]: a two-layer network (Linear $64 \rightarrow 32$ with tanh, then Linear $32 \rightarrow 1$) produces a scalar relevance score for each timestep, which is normalised via softmax and used to form a weighted sum over the time dimension, yielding a single 64-dimensional summary vector. This vector is passed through a two-layer classification head (Linear $64 \rightarrow 48$, ReLU, 20% dropout, Linear $48 \rightarrow 1$) that produces the raw impairment logit, converted to a probability $\hat{p} \in [0, 1]$ via sigmoid.

Handling class imbalance. Error-preceding windows are structurally rare: the mean positive rate across participants is approximately 9.5%. Two complementary interventions address this imbalance at the data level and the loss level, respectively. At the data level, we apply a session-aware variant of SMOTE [5]: synthetic positive windows are generated by interpolating between $k=5$ nearest neighbours, the default recommended [5], identified in a PCA-reduced representation of the flattened raw signal space, with the number of retained components set adaptively to $\min(n_{\text{pos}} - 1, D, 50, \max(2, \lfloor n_{\text{pos}}/5 \rfloor))$, where n_{pos} is the number of positive training windows and D is the flattened window dimension; this caps the projection at 50 components while ensuring it never exceeds the number of available positive examples. The target positive-class ratio is set to 0.30, a value validated on the inner validation fold (25% of training drivers, stratified by positive-rate presence, held out for calibration and early stopping) as the value that maximised AUROC without introducing excessive synthetic variance; lower ratios left the model undertrained on the minority class, while higher ratios degraded generalisation. To preserve temporal coherence, interpolation is restricted to window pairs from the same driving session that are separated by at least $L=45$ s; no synthetic window may span or mix sessions. At the loss level, we adopt focal loss [20] with concentration parameter $\gamma=2$, the value used in the original formulation for detection tasks with comparable imbalance ratios, which down-weights confident predictions and concentrates the training signal on ambiguous windows near the decision boundary. Each training sample is additionally weighted by its composite risk score, normalised by the mean positive-class score across the training fold, so that windows preceding high-severity events such as collisions exert a proportionally stronger influence on the model than those preceding minor incidents; normalisation by the mean rather than the maximum preserves the full severity gradient across folds where the most severe event types are rare, while keeping effective per-sample weights in a stable range. Safe windows carry unit weight.

Optimisation and regularisation. Each model instance is trained with the Adam optimiser ($\eta=5 \times 10^{-4}$, $\lambda_{\text{wd}}=10^{-4}$) for up to 100 epochs with a cosine annealing learning-rate schedule ($\eta_{\text{min}}=5 \times 10^{-6}$) and gradient clipping at norm 1.0. Training is halted early when validation-fold AUROC fails to improve for 20 consecutive epochs. Two data-space regularisers limit overfitting to session-specific patterns: additive Gaussian jitter on the input features ($\sigma=0.01$) and random per-feature amplitude scaling (uniform on $[0.8, 1.2]$). All augmentation scales were chosen conservatively relative to the z-scored input range and validated on the inner fold to ensure they perturb the kinematic representation without distorting the temporal signatures the model is trained to detect.

Ensemble and calibration. To reduce sensitivity to random initialisation, $K=10$ independent model instances are trained per fold; their sigmoid outputs are averaged to form the ensemble score. This ensemble size follows the work done in [18], which shows that predictive uncertainty and accuracy gains

plateau beyond five to ten members, making larger ensembles unnecessary given the computational cost of LOPO-CV. Raw neural network outputs are typically poorly calibrated [9]; the ensemble score is therefore recalibrated with Platt scaling [24], fitting a logistic transform on the inner validation fold. Calibration is accepted only when the fitted weight is positive and bounded ($0 < w \leq 20$); otherwise the raw ensemble score is passed through unchanged, preventing degenerate rescaling in folds where the validation fold is single-class or the recalibrator would invert the score ordering. The fitness-to-drive index delivered to the end user is $f = 1 - \hat{p}$, so that f close to 1 reflects a driver in full control and f close to 0 signals elevated impairment risk.

Hyperparameter summary. Table 1 consolidates all architectural and training hyperparameters for reproducibility.

Table 1: KinTCN architecture and training hyperparameters.

Component	Parameter	Value
Architecture	Residual blocks	5
	Dilations	1, 2, 4, 8, 16
	Channels	$36 \rightarrow 32 \rightarrow 64$ (held)
	Receptive field	63 timesteps
	Attention hidden dim	32
Regularisation	Block dropout (Dropout1d)	0.10
	Head dropout	0.20
	Input jitter σ	0.01
Training	Optimiser	Adam
	Learning rate η	5×10^{-4}
	Weight decay λ	10^{-4}
	LR schedule	Cosine annealing ($\eta_{\min}=5 \times 10^{-6}$)
	Max epochs / patience	100 / 20
	Batch size	64
	Gradient clip (norm)	1.0
Imbalance	Focal loss γ	2.0
	SMOTE target positive rate	0.30
	SMOTE neighbours k	5
	SMOTE PCA cap	50 components
Ensemble & calibration	Ensemble size K	10
	Calibration	Platt scaling (val fold)
	ECE bins	10 equal-width

5 KinTCN Evaluation

All results presented in this work are reported under the strict Leave-One-Participant-Out Cross-Validation (LOPO-CV) protocol: in each fold, one participant is held out entirely as the test set, and all preprocessing, SMOTE generation, and model training are performed exclusively on the remaining participants. This is the strictest available cross-participant evaluation protocol and directly simulates deployment on a driver the system has never encountered. The primary performance metric is the area under the ROC curve (AUROC), which measures the model’s ability to rank impaired windows above safe ones independently of any operating threshold; calibration quality is additionally assessed through Expected Calibration Error (ECE) [9], computed using 10 equal-width probability bins over $[0, 1]$. We do not report Brier score: under the severe class imbalance present in this dataset (mean positive rate $\approx 9.5\%$), a trivial model that assigns the base rate to every window achieves a near-optimal Brier score without any discriminative ability, making it an uninformative and potentially misleading metric for this setting. All aggregate metrics are accompanied by 95% confidence intervals obtained via non-parametric bootstrap ($N = 2\,000$ resamples).

Not all participants contribute equally to this evaluation. Those with fewer than five positive windows *or* a positive rate below 5% in their held-out fold provide insufficient signal for reliable AUROC estimation and are designated as *audit drivers*. For this group, the relevant safety question shifts from discrimination to false alarm control: rather than AUROC, we report the false positive rate (FPR) at a fixed operating threshold, quantifying how often the model raises an impairment alert for a driver who committed no safety-critical error.

Before reporting main results, we present a systematic ablation over the two key windowing hyperparameters: the lookback duration L and the labelling horizon H . To ensure a fair comparison across all configurations, metrics are computed on the subset of participants present in every (L, H) cell, yielding a common cohort of 16 drivers. The configuration maximising AUROC on this common cohort is selected and used for all results that follow.

5.1 Lookback and Horizon Ablation

We conduct a systematic grid search over two windowing hyperparameters: the lookback duration $L \in \{30, 45, 60\}$ s, which controls how much kinematic history the model receives as input, and the labelling horizon $H \in \{10, 15, 20\}$ s, which defines the future window within which a safety-critical error must occur for a window to be labelled as positive. This yields nine configurations evaluated under the same LOPO-CV protocol described above. To ensure strict comparability, all cells are evaluated on the common subset of 16 participants present across every configuration; participants excluded from any single cell due to insufficient session length are omitted from all cells. AUROC on this common cohort is used as the sole selection criterion. AUPRC is excluded from selection because its magnitude is directly coupled to the positive rate, which

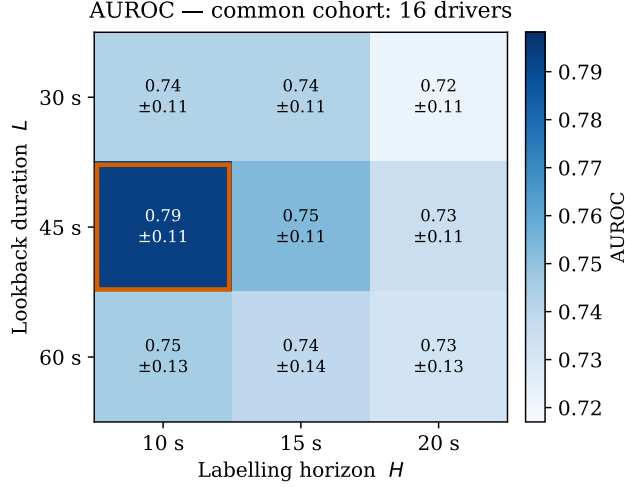


Figure 1: AUROC as a function of L and H , evaluated on the 16-driver common cohort under LOPO-CV. The orange rectangle marks the best-performing configuration ($L=45$ s, $H=10$ s).

increases mechanically with H ; using it would systematically bias the search towards larger horizons regardless of true discriminative gain. ECE measures calibration quality, which is adjusted post-hoc via Platt scaling independently of (L, H) ; it therefore reflects a correctable property of the raw scores rather than an intrinsic characteristic of the configuration, and is not an appropriate basis for model selection.

Figure 1 presents the resulting 3×3 AUROC heatmap. Two consistent patterns emerge from the grid. First, AUROC peaks at $L=45$ s and degrades in both directions. The improvement over $L=30$ s ($+0.051$ at $H=10$) suggests that 30 seconds of kinematic context is insufficient to capture the full behavioural signature of impairment onset. The degradation at $L=60$ s, accompanied by a marked increase in cross-participant variance ($\sigma=0.134$ vs. 0.107 at $H=10$), indicates that very long lookback windows introduce noise: as session length constrains the number of valid windows per participant, training becomes less stable and generalisation suffers. Second, AUROC decreases monotonically as H increases, with the effect consistent across all lookback values. This is consistent with the hypothesis that the discriminative link between the kinematic state at time t and a subsequent error weakens as the prediction horizon widens: errors occurring 15–25 s after the gap are more likely to arise from situational factors unrelated to the impairment state captured in the lookback window.

Based on this analysis, we select $L=45$ s and $H=10$ s as the operating configuration for all subsequent experiments. This setting achieves **AUROC** = **0.79 ± 0.11** on the common ablation cohort and represents the best trade-off between contextual richness and prediction precision.

Model comparison ($L = 45 \sim s$, $H = 10 \sim s$, $G = 5 \sim s$)

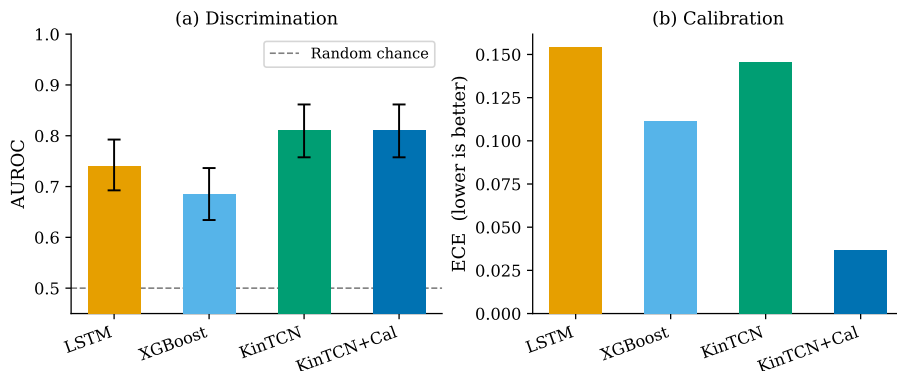


Figure 2: AUROC (left) and ECE (right) for all models under the selected configuration ($L=45$ s, $H=10$ s). Error bars denote 95% bootstrap confidence intervals. The dashed line marks chance level.

5.2 Main Results

KinTCN+Cal achieves an AUROC of 0.810 [0.758, 0.862] across the 18 evaluable participants, representing gains of +0.071 over LSTM and +0.126 over XGBoost (Figure 2). The separation is consistent across the full operating range, as visible in the ROC curves of Figure 3 (left), and is not an artefact of pooling: at the per-participant level, KinTCN+Cal outperforms LSTM in 15 of 18 cases, with a mean gain of $+0.071 \pm 0.079$ that reaches statistical significance under the Wilcoxon signed-rank test ($W=14$, $p=0.001$, Figure 4, left). The advantage over XGBoost is equally consistent: KinTCN+Cal outperforms XGBoost in 15 of 18 cases ($W=12$, $p=0.001$, Figure 4, right). This demonstrates that KinTCN’s dilated receptive field and attention-based pooling extract more discriminative temporal patterns than both flat feature aggregation (XGBoost) and recurrent sequence modelling (LSTM).

Under the severe class imbalance present in this dataset, ROC curves can overstate practical utility by crediting true negatives that are trivially abundant. The Precision-Recall curves (Figure 3, right) provide a more demanding view: KinTCN achieves a mean average precision of 0.409, against 0.253 for XGBoost and 0.326 for LSTM, all substantially above the no-skill baseline.

Calibration tells a complementary story. KinTCN and KinTCN+Cal are identical in AUROC, yet their ECE values diverge sharply: 0.146 before scaling and 0.037 after, a 75% reduction. Among the baselines, XGBoost achieves a native ECE of 0.112, while LSTM reaches 0.154: both considerably higher than KinTCN+Cal and consistent with the observation that more expressive models tend to produce less well-calibrated raw scores unless explicitly regularised for probability estimation.

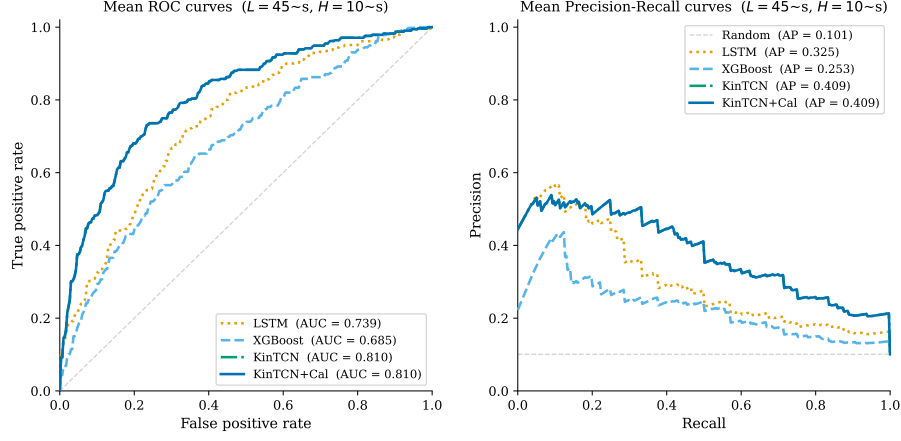


Figure 3: Mean ROC curves (left) and Precision-Recall curves (right) for all models ($L=45$ s, $H=10$ s). The dashed horizontal line in the PR plot marks the no-skill baseline (positive rate).

Feature Importance. Among the four kinematic channels, longitudinal speed emerges as the dominant predictor ($\Delta\text{AUROC} = +0.104 \pm 0.119$), with steering torque and steering wheel angle contributing comparably ($+0.058$ and $+0.057$, respectively) and lateral acceleration providing the smallest but still positive contribution ($+0.053 \pm 0.081$, Figure 5). That all four signals yield a strictly positive mean importance rules out the possibility that any channel is acting as noise; the model exploits each input without redundancy harming performance. The prominence of speed is consistent with its role as a high-level regulator of driving behaviour: distracted drivers tend to reduce speed variability and lower their mean speed as a compensatory strategy, a pattern the 45-second lookback window is well-positioned to detect.

Safe Driver Audit. A model that discriminates well but raises frequent false alarms on safe drivers would be of limited practical value. To assess this, the 22 participants below the eligibility threshold were each evaluated in a dedicated held-out fold, and the mean calibrated impairment probability assigned to their error-free windows was recorded. As Figure 6 shows, every audit participant sits comfortably below the 0.5 alert threshold, with a group mean of 0.054 (range [0.023, 0.081]). Crucially, this holds for both subgroups: participants with no recorded errors throughout the session, and those who produced between one and four — too few for AUROC estimation but enough to confirm that impairment episodes did occur. The absence of systematic false alarms across the full audit cohort suggests that the model has not learnt to flag any kinematic deviation as impaired, but rather responds selectively to the patterns that genuinely precede safety-critical errors.

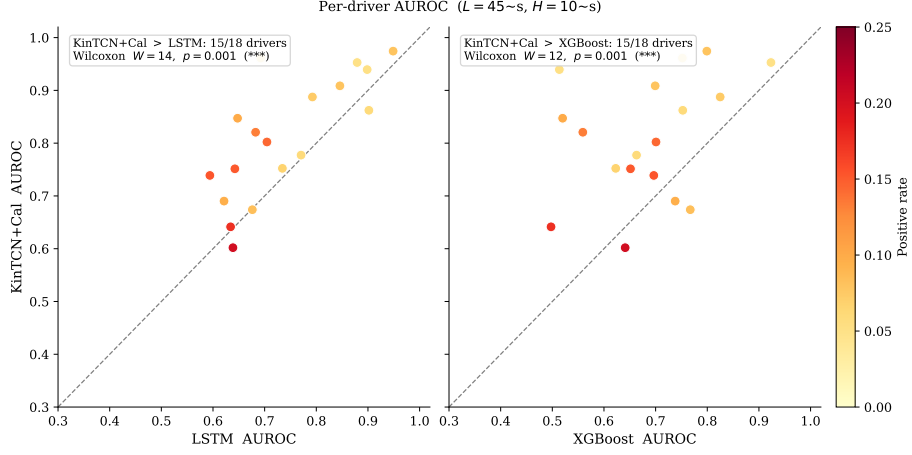


Figure 4: Per-participant AUROC: KinTCN+Cal vs. LSTM (left) and vs. XGBoost (right). Colour encodes positive rate. Points above the diagonal indicate KinTCN+Cal superiority.

6 Discussion

KinTCN’s ability to rank impaired driving windows above safe ones (AUROC 0.810 under a protocol that excludes the test participant from every preprocessing and training step) is grounded in a well-established psychophysical mechanism rather than in an unexplained statistical regularity. Engström et al.’s cognitive control hypothesis [8] predicts that increased cognitive load reduces the bandwidth of the driver’s closed-loop vehicle control, producing kinematic perturbations that precede overt errors. The permutation importance results in Figure 5 are consistent with this account at the level of individual signals. Speed emerges as the dominant predictor ($\Delta\text{AUROC} = +0.104$), which aligns with the compensatory speed-reduction behaviour documented in naturalistic distraction studies [28, 27]: cognitively loaded drivers tend to attenuate speed variability and lower mean cruise speed as a strategy for reducing the information-processing demands of the driving task. This signature unfolds over tens of seconds, which explains why the 45-second lookback window outperforms the 30-second one: the compensatory response is a gradual, sustained shift in vehicle control, not a transient event. Steering torque and steering angle contribute comparably ($+0.058$ and $+0.057$), consistent with steering entropy theory [21, 4] and the selective sensitivity of steering reversal rates to cognitive, as opposed to visual, distraction demonstrated by Kountouriotis et al. [15]. Lateral acceleration provides the smallest but still positive contribution, which is plausible given that lateral dynamics integrate both speed and steering inputs and therefore carry partially redundant information relative to those two channels. That all four contributions are strictly positive rules out the possibility that any

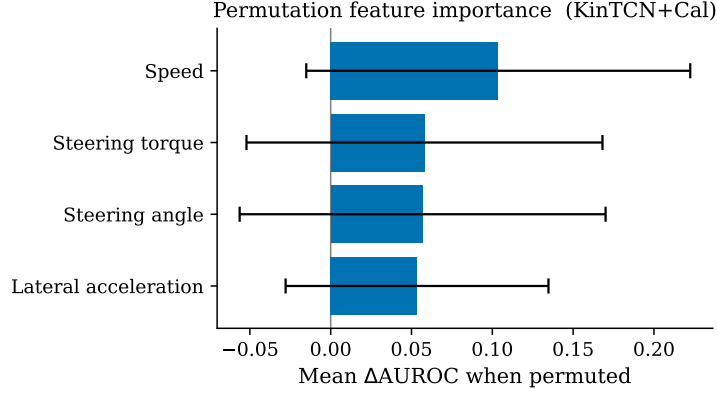


Figure 5: Permutation importance: mean AUROC drop when each signal is independently permuted across 10 repetitions per fold. Error bars denote standard deviations across folds.

channel acts as noise under this evaluation protocol.

The fact that the model’s discriminative ability peaks at $H=10$ s and degrades monotonically with longer labelling horizons has a straightforward interpretation: the kinematic state at time t is most informative about errors that will occur in the near future. Errors occurring 15–25 s after the observation gap are increasingly likely to arise from situational factors (sudden road geometry changes, unexpected lead-vehicle behaviour) that are orthogonal to the driver’s momentary impairment state. This decay in predictive relevance with horizon is consistent with the theoretical position that kinematic signals are indicators of impairment *state*, not predictors of discrete error *events* [6], and it provides empirical support for framing the problem as continuous state scoring rather than event forecasting.

6.1 The Fitness-to-Drive Index as an Actionable Output

A discrimination metric such as AUROC establishes that the model can rank windows correctly, but it says nothing about whether the raw probability estimates are meaningful. For a fitness-to-drive index to be deployable (whether as a driver alert, an insurer risk metric, or a fleet management signal) the output must be probabilistically meaningful: a score of 0.8 should correspond to an empirically higher impairment rate than a score of 0.4. Platt scaling [24] reduces the Expected Calibration Error from 0.146 to 0.037, a 75% reduction, bringing the calibrated score into a range that supports threshold-based alerting. The practical significance of this is visible in the audit driver results: every one of the 22 participants below the AUROC eligibility threshold receives a mean calibrated impairment probability well below the 0.5 alert threshold (group mean 0.054, range [0.023, 0.081]), including those who committed between one and

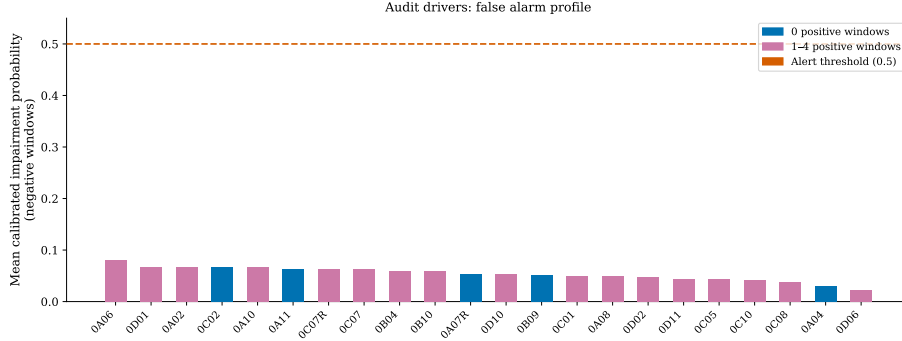


Figure 6: Mean calibrated impairment probability on error-free windows for each of the 22 audit participants. The dashed line marks the 0.5 alert threshold. Colour distinguishes participants with zero positive windows (blue) from those with one to four (pink).

four incidents. This low false alarm rate on the audit cohort is not a consequence of the model assigning uniformly low scores, as it responds selectively to the kinematic patterns that precede safety-critical events in the held-out evaluable participants. Together, the calibration and audit results suggest that the index behaves as intended: it is near-zero during unambiguously safe driving and elevates selectively when kinematic patterns consistent with impairment are present.

6.2 Limitations and boundary conditions

Several limitations bound the scope of the conclusions that can be drawn from these results and should guide the design of follow-on work.

Simulator validity and ecological generalisability. Controlled simulator studies are the only feasible setting in which safety-critical events can be observed at sufficient frequency and under reproducible conditions: inducing collisions, red-light violations, and panic braking episodes in real traffic is neither safe nor ethically permissible. The simulator setting is therefore not a methodological compromise but a necessary precondition for obtaining the labelled data that weak supervision requires. Crucially, KinTCN is designed with deployment in mind: the cross-participant evaluation protocol and the per-session normalisation strategy ensure that the model generalises to drivers it has never encountered, which is precisely the condition it faces at first deployment on a new vehicle. Once deployed, the system is naturally positioned to transition towards personalised continuous learning, refining its impairment estimates from the accumulating kinematic history of each individual driver and progressively reducing reliance on population-level priors. Validating this online adaptation loop on naturalistic driving data is an important direction for future work.

Sample size and positive rate heterogeneity. Forty participants were recruited, but only 18 met the eligibility threshold for AUROC estimation in their held-out fold. It is important to note that the remaining 22 participants are not simply discarded: they appear in the training folds of all other participants and therefore contribute to the learned representations, and they are evaluated in a dedicated audit protocol that assesses false positive rate on their error-free windows, which is the safety-relevant question for drivers whose incident rate is too low to support discrimination estimation. This dual role is a strength of the design rather than a limitation: the audit cohort provides evidence that the model does not over-alert on safe drivers, which is a prerequisite for realistic deployment. Nevertheless, the 18-participant evaluable cohort produces wide confidence intervals on AUROC (± 0.052), and the per-participant variance visible in Figure 4 indicates that individual differences in baseline driving style and distraction response are substantial. Future work should increase the sample size and, ideally, stratify recruitment to ensure a minimum positive rate per participant, which would both narrow the confidence intervals and reduce the proportion of participants relegated to audit-only evaluation.

Weak supervision and label noise. Weak supervision is not a limitation specific to this work: it is the only honest form of supervision available without the physiological sensors that the kinematic-only approach is designed to replace. A system trained on clean physiological ground truth would, by construction, require those very sensors at deployment, defeating the purpose of the index. The label noise introduced by using safety-critical events as a proxy for impairment state acts as a conservative bias on reported performance: the model must learn to score a latent state from a signal that is only noisily correlated with it, so the AUROC of 0.810 should be interpreted as a lower bound on the true discriminability of kinematic impairment signatures rather than as a ceiling. The severity-weighting scheme further mitigates this by concentrating the training signal on the least ambiguous cases, i.e., windows preceding high-consequence events that are least likely to be situationally driven.

Distraction type specificity. Cognitive distraction is not merely one impairment type among many: it is the one that existing detection systems handle least well. Camera-based gaze monitors fail precisely because the cognitively distracted driver’s eyes remain on the road; physiological monitors are not deployed at scale; and the kinematic signature of cognitive load, a gradual reduction in closed-loop control bandwidth, is subtler and harder to detect than the gross lane departures or head drops associated with drowsiness or visual distraction. Demonstrating that KinTCN achieves AUROC 0.810 for this hardest case is therefore a stronger result than equivalent performance on more visually salient impairment modes would be. Drowsiness and alcohol intoxication, which produces stronger and clearer kinematic degradation [27], are plausible targets for our architecture with appropriate training data, and represent a natural extension rather than an open question about the viability of our approach.

7 Conclusion

This work addressed the problem of driver impairment detection as continuous state scoring rather than discrete error prediction, a reframing that is both theoretically motivated and practically necessary. The correct question for a deployable safety system is not whether a driver will make an error in the next N seconds, but whether the driver is currently in a state associated with elevated error risk. KinTCN operationalises this framing into a fitness-to-drive index derived exclusively from four kinematic signals available in any modern vehicle: speed, steering wheel angle, steering torque, and lateral acceleration.

Evaluated under LOPO-CV with strict anti-leakage normalisation, i.e., the only protocol that honestly simulates deployment on an unseen driver, the calibrated KinTCN achieves an AUROC of 0.81 [0.76, 0.86] and an Expected Calibration Error of 0.037 after Platt scaling, representing gains of +0.071 in discrimination and 75% in calibration quality over the best recurrent baseline. The permutation importance analysis confirms that all four kinematic channels contribute positively and consistently with the psychophysical mechanisms established in the driving psychology literature: speed dominates as a marker of compensatory load-reduction behaviour, whilst steering torque and angle reflect the irregularity in closed-loop vehicle control predicted by steering entropy theory. The false positive audit on the remaining 22 participants, i.e., those whose incident rate was too low for discrimination estimation, shows that every driver sits well below the 0.5 alert threshold, with a group mean calibrated impairment probability of 0.054, confirming that the index does not over-alert on safe driving.

The broader methodological contribution is a demonstration that cross-participant generalisation and probabilistic calibration are jointly necessary conditions for a deployable fitness-to-drive index, and that within-subject evaluation protocols systematically obscure both. Future work should validate the online adaptation of the index on naturalistic driving data, extend the architecture to additional impairment types such as drowsiness and alcohol intoxication, and increase sample size to narrow the confidence intervals that the current simulator study, by necessity, leaves wide. KinTCN and the labelled dataset are publicly available to support this line of work [1].

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